

Nathan Potraz

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EXPERIENCE

Hardware Specialist & Small Business Owner

Self-Employed • June 2022 – Present • Oceanside, CA

- Engineered over 100 custom-tuned competitive gaming controllers, earning an average rating of 5/5.
- Work alongside clients, establishing their requirements and tuning their controllers to meet their specifications.
- Handle all aspects of the business – from supply chain, to circuit-board soldering, firmware flashing, customer service, and tech support.
- Contributed to the open-source motherboard project with board testing/measurements, and documentation.

PROJECTS

Devlog

devlog.ntpotraz.dev/ • October 2025 – Present

- A journaling web application built using **TypeScript React** and a **Golang HTTP server and API**.
- Features simple and fast journal entry **creation, sharing, and editing** utilizing **SQLite hosting** with Turso.
- Integrated Clerk for **secure user authorization**, supporting GitHub and Google SSO as well as email/password.
- Streamlined deployment using **Docker and GitHub Actions** for automated CI/CD to **Google Cloud Run**.

Pacific Islander Community Health & Resource Mobile Application

ntpotraz.github.io/PIC-Health-App/ • August 2024 – December 2024

- Developed a resource and event calendar **React Native mobile application** for underrepresented communities.
- **Lead a team of 5 developers** to work alongside our sponsor to **plan, design, and develop** our Capstone project.
- Set up GitHub repo, Expo project, and assisted my dev team with their environments and using version control.
- Integrated **Google API** to fetch and render **community events** for users from the admin Google Calendar.

Japanese-English Translator

github.com/nthpotraz/cs421-translator/ • March 2024 – May 2024

- Engineered a **C++ natural language processor** using a **DFA Scanner** to tokenize complex linguistic inputs.
- Built a **Recursive Descent Parser** to enforce strict grammar rules, mirroring simulation logic structures.
- Developed a translation engine using **Hash Maps** for efficient dictionary lookup and token reordering.
- Integrated **File I/O streams** with **custom error handling** to ensure high-fidelity output and debugging.

Custom DNS Resolver Simulation

github.com/Saltair1/NetworkPythonProject/ • October 2023 – November 2023

- Developed a **custom Python protocol over UDP** to simulate **Domain Name System** infrastructure.
- Engineered **local caching logic** to validate record freshness by comparing timestamps against **TTL expiration**.
- Integrated **Pandas** to visualize resource record tables, providing **real-time monitoring** of the resolver state.
- Automated **server configuration** using JSON files to manage static records across multiple **authoritative nodes**.

EDUCATION

B.S in Computer Science

California State San Marcos • San Marcos, CA • 3.7 GPA

Relevant Coursework: OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, Computer Security, AI, Operating Systems, AGILE/SCRUM.

SKILLS

Languages: Golang, Python, Java, C++, TypeScript, React, SQL

Tools: Linux, Docker, CI/CD, Git, Linux, Agile, AWS, GCP